Measurement 252 (2025) 117329



Contents lists available at ScienceDirect

Measurement

journal homepage: www.elsevier.com/locate/measurement





A real-time recognition and distance measurement method for underwater dynamic obstacles based on binocular vision

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ARTICLE INFO

Keywords:
Underwater refraction model
Image enhancement
Obstacle detection
Stereo matching
Semi-global block matching (SGBM)

ABSTRACT

Real-time and accurate distance measurement of dynamic obstacles is essential for underwater tasks such as exploration and structural maintenance to prevent collisions. However, recognizing and measuring underwater obstacles pose significant challenges due to limited visibility, water turbidity, and complex lighting conditions. These factors obscure obstacle features, distort visual information, and reduce the accuracy of obstacle identification and distance measurement. To address these challenges, an accurate and real-time method is proposed for recognizing and measuring underwater dynamic obstacles using binocular vision in this paper. Firstly, an innovative calibration model for underwater cameras is developed to eliminate the distortion caused by refraction between water, the waterproof housing, and air, thus achieving precise camera parameters. Secondly, an image enhancement method combining the dark channel prior (DCP) and adaptive clipped contrast limited histogram equalization (ACCLAHE) is employed to improve image contrast and detail visibility. Finally, YOLOv5s is employed for obstacle detection, together with an improved semi-global block matching (SGBM) algorithm accelerated by multi-threaded parallel computing to enhance matching speed. This approach enables rapid and accurate recognition and distance measurement of underwater dynamic obstacles. The effectiveness of the proposed algorithm is validated through underwater experiments using both qualitative and quantitative metrics.

1. Introduction

Various underwater robots have been deployed in ocean research areas such as marine rescue, underwater exploration, and pipeline inspection. However, dynamic obstacles such as fishing nets, sunken ships, and other objects pose significant threats to underwater platforms. Therefore, accurately recognizing and assessing these obstacles is crucial for assisting underwater vehicles in avoiding collisions.

Currently, various measurement technologies have been proposed and implemented in sensor systems [1,2]. Among these, acoustic sensors are the most widely used in underwater measurement applications. However, their inherent limitations in sensitivity, resolution, and operational range render them unsuitable for real-time and high-accuracy measurements at short distances. To address these limitations, vision-based stereo matching has emerged as an effective solution for achieving precise measurements at close range, offering high sensitivity and resolution. By generating disparity and depth maps, stereo matching enables accurate distance measurement of obstacles in various environments. Nevertheless, underwater stereo matching faces significant challenges due to the inherent complexities of the underwater environment. Factors such as water refraction, reduced contrast, and low-light conditions severely degrade the performance of visual information capture and processing.

To address the aforementioned challenges, numerous studies have concentrated on pre-processing techniques, including image enhancement and camera modeling. However, limited visibility and turbidity in the harsh underwater condition lead to low-quality images, which reduces the effectiveness of most existing image enhancement techniques. Furthermore, the geometric distortion in camera imaging due to refraction and scattering in water complicates the acquisition of accurate calibration parameters. Therefore, this paper presents a real-time recognition and distance measurement approach for underwater dynamic obstacles using binocular vision. The proposed approach primarily comprises an underwater binocular camera calibration module, an underwater image enhancement module, a YOLOv5s based obstacle detection module, and an obstacle distance measurement module. In the

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https://doi.org/10.1016/j.measurement.2025.117329
Received 6 November 2024; Received in revised form 12 March 2025; Accepted 16 March 2025
Available online 20 March 2025
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underwater binocular camera calibration module, a novel calibration model is developed to correct the complex distortion caused by refraction between water, the waterproof housing, and air. In the underwater image enhancement module, an image enhancement method combining dark channel prior (DCP) with adaptive clipped contrast limited histogram equalization (ACCLAHE) is employed to enhance image contrast and details while minimizing color distortion. The DCP algorithm effectively mitigates color distortion commonly observed in underwater images, whereas the ACCLAHE algorithm dynamically adjusts the number of sub-images and clipping limit to enhance both contrast and detail. Additionally, YOLOV5s is employed for dynamic obstacle detection. In the obstacle distance measurement module, an enhanced multi-threaded semi-global block matching (SGBM) algorithm, which introduces the MAD-Census transform into multi-threaded parallel computation, is developed to expedite the matching process and improve the accuracy of dynamic obstacle distance estimation in underwater environments. Experimental results indicate that the proposed algorithm effectively recognizes underwater obstacles and measures their distances in real time.

The key contributions of this paper can be outlined as follows:

(1) This paper proposes a unified framework that can simultaneously perform underwater image enhancement, underwater obstacle detection, and underwater obstacle distance measurement for real-time collision avoidance applications in underwater environments. The framework is designed as a lightweight model for running on embedded processors with limited computational resources. Underwater experiments illustrate the capability of the proposed framework in achieving real-time distance measurement of dynamic obstacles in various underwater scenarios.
(2) A novel underwater camera calibration model is proposed to eliminate additional distortion caused by refraction between water, the waterproof housing, and air, thereby enabling the acquisition of accurate camera parameters.
(3) An image enhancement method based on DCP and ACCLAHE is designed to enhance image contrast and details while avoiding color distortion. The DCP algorithm effectively mitigates the color distortion, whereas the ACCLAHE algorithm dynamically adjusts both the quantity of sub-images and the clipping limit values to enhance image contrast and details. This approach provides high-quality images for subsequent distance measurement tasks.
(4) A new stereo matching method is proposed based on an improved multi-threaded optimized SGBM algorithm. The SAD cost in the traditional SGBM algorithm is replaced with the MAD-Census transform, and the multi-threaded parallel computation is adopted to improve computational efficiency. By utilizing the disparity values obtained from the improved SGBM algorithm, precise measurement of the relative distances between the camera and obstacles is achieved.

The following sections of this paper are organized as follows: Section 2 reviews the related work, while Section 3 provides an overview of the fundamental principles of the proposed obstacle distance measurement system. In Section 4, the specific details of the proposed framework are described. Specifically, Section 4 comprises an underwater binocular camera calibration module in Section 4.1, a YOLOv5s obstacle detection module in Section 4.2, an underwater image enhancement module in Section 4.3, and a distance measurement module in Section 4.4. In Section 5, the experimental results are presented. Lastly, the conclusions are detailed in Section 6.

2. Related work

2.1. Underwater camera calibration

Although some specially designed underwater cameras with built-in waterproof features can be directly used in water without additional protective casings, such options are often prohibitively expensive. Standard cameras that are widely used in industry can be sealed within waterproof housings to capture underwater images. Consequently, current research primarily focuses on the use of standard cameras in underwater environments. However, during the underwater image capture process, light passes through multiple media before reaching the imaging plane, resulting in significant refraction. Camilo et al. [3] proposed an underwater camera model that does not account for refraction effects. However, neglecting these effects can lead to substantial deviations in image quality and measurement accuracy.

To address the effects of refraction, some studies have adopted an auxiliary calibration board to calibrate camera parameters [4,5]. Although this method is easy to implement, it does not fully address the refraction problem and may result in considerable calibration errors in underwater environments. Some researchers interpret the underwater refraction as a type of lens distortion, and regard pixel offset errors as refraction-related errors arising from lens distortion [6]. Therefore, the effects of refraction can be mitigated by calibrating camera parameters and correcting image distortion. However, the influence of refraction varies significantly across different pixels, making it challenging to correct the entire image with uniform distortion parameters. Calibration algorithms based on physical models have also been developed for underwater imaging. The non-dominated sorting genetic algorithm II (NSGA-II) has been proposed for flexible omnidirectional tilted cameras, aiming to enhance the effectiveness of underwater measurement [7]. To effectively reconstruct three-dimensional scenes, Ruihang Miao et al. [8] proposed a single viewpoint (SVP) underwater camera model to correct lens and refraction distortions occurring from water to air. However, the SVP model does not solve the refraction caused by the waterproof housing. In this paper, a new refraction model that considers the thickness of the waterproof housing and the complicated refractions is introduced for underwater cameras. This method aims to facilitate accurate calibration of camera parameters.

2.2. Underwater image enhancement

Image enhancement is a crucial step in improving the visual quality of underwater images. Image enhancement methods can be classified into three distinct categories: physical model-based methods, non-physical model-based methods, and deep learning-based methods. Physical model-based approaches employ underwater optical imaging models to recover the true colors and details of images through rigorous physical modelling, as demonstrated by techniques such as dark channel prior (DCP) [9]. Yu et al. [10] applied a DCP-based color correction method to enhance image contrast. W. Zhang et al. [11] suggested an approach to improve underwater image quality through various operations, including minimizing color loss, image fusion, and color balancing. However, due to the complexity of underwater environments, these methods face challenges in obtaining accurate physical parameters. Non-physical model-based methods aim to enhance image quality by adjusting pixel values and their distributions. These methods include traditional CLAHE [12], Retinex [13], and others. To address issues of image blurring and color deviation, Gao et al. [14] introduced a retina-guided image enhancement method for deep-sea fish research. Zhang et al. [15] combined color correction with dual interval contrast enhancement methods to improve underwater image quality. Although these methods can restore underwater images, they often face challenges of over-enhancement or under-enhancement in harsh underwater conditions. Deep learning-based methods learn degradation characteristics of underwater images from large datasets, thereby producing high-

quality enhancement results. Zhou et al. [16] introduced a hybrid contrastive learning regularization approach that combines an adaptive mixed attention module with a detail restoration branch, thereby enabling efficient feature extraction and texture detail recovery. Wang et al. [17] introduced a reinforcement learning-based method for underwater image enhancement. This approach systematically integrates multiple image enhancement algorithms to progressively optimize the enhancement process. While these methods demonstrate strong performance in restoring underwater images, they often rely heavily on specific underwater datasets. Additionally, some approaches neglect the physical principles underlying underwater imaging degradation, leading to enhancement images that may not accurately reflect reality. In this work, an image enhancement method based on DCP and ACCLAHE is proposed to simultaneously enhance image contrast and detail while avoiding color distortion.

2.3. Deep learning based object detection

Object detection focuses on identifying objects of interest in an image and classifying their categories. Recently, deep learning approaches have been extensively utilized in object detection due to their strong robustness and generalization ability. Based on the generation of candidate regions, object detection frameworks that employ deep learning can be divided into two main approaches: the two-stage method and the one-stage method. Two-stage algorithms encompass two essential steps. The first step involves proposing candidate regions of interest, followed by a second step that refines the detected regions to obtain the final detections with their bounding boxes. Two-stage methods such as R-CNN [18] and Faster R-CNN [19] can produce high detection accuracy. However, their slow processing speed renders them unsuitable for real-time applications.

To improve detection speed, researchers have proposed one-stage methods, including the YOLO series algorithms. Redmon et al. [20] introduced YOLOv1 to predict both the locations and classifications of targets. Building upon this, YOLOv2 was developed by incorporating Batch Normalization into YOLOv1 to accelerate training speed [21]. Subsequently, Redmon designed YOLOv3 [22] by introducing Darknet53, a network structure with superior classification capabilities. After YOLOv3, numerous improved versions of the YOLO series have emerged. For instance, YOLOv5 [23] model achieves high precision and rapid real-time object detection through a streamlined and compact architecture, as well as optimized feature extraction. This makes it particularly suitable for deployment on edge devices. YOLOv6 [24] enhances the series by incorporating advanced feature pyramid networks to improve detection accuracy. YOLOv7 [25] introduces innovative training strategies and a more efficient network structure. YOLOv8 [26] adopts an improved network structure based on cross-stage partial network (CSPNet). YOLOv9 [27] introduces dynamic convolution and multi-scale feature fusion to strengthen detection capabilities. Meanwhile, YOLOv10 [28] incorporates squeeze-and-excitation (SE) and convolutional block attention module (CBAM) to improve feature map effectiveness. YOLOv11 [29] combines transformer architectures with convolutional neural networks (CNNs) to achieve an effective integration of global context and local features. However, these special designs after YOLOv5 increase both computational load and model complexity, with a slight reduction in inference speed. Among these one-stage methods, YOLOv5s strikes a balance between detection speed and accuracy. Therefore, this paper adopts YOLOv5s for real-time obstacle detection.

2.4. Stereo matching

In underwater environments, distance measurement encounters numerous challenges, including light attenuation, light scattering, and dynamic changes in obstacle positions. These factors pose challenges to the accurate estimation of the distance between vehicles and obstacles. To tackle these difficulties, numerous researchers have proposed stereo matching technology for underwater distance measurement. Stereo matching is mainly categorized into feature matching and region matching. Traditional feature matching algorithms include SURF [30], SIFT [31], and ORB [32]. Although these methods perform well across various application scenarios, they exhibit limited robustness in handling scale variations, illumination changes, and viewpoint transformations. To address these limitations, deep learning-based feature matching algorithms have been proposed in recent years. ASTR (adaptive scale transform registration) [33] enhances feature matching capabilities at different scales through adaptive scale transformations, while SFD2 (sparse feature detection 2) [34] improves matching efficiency and reliability by optimizing the feature point detection and description processes. Region matching algorithms, such as Block Matching (BM) [35], SGBM [36], and SGM [37], are widely utilized in stereo matching. The SGM algorithm is renowned for its exceptional capability in approximating optimization through two-dimensional MRF (markov random field) inference. The SGBM algorithm improves upon the SGM algorithm by incorporating the preprocessing and post-processing techniques of the BM algorithm, thereby enhancing both matching precision and real-time performance. However, these algorithms still face high computational complexity and limited adaptability, particularly when processing high-resolution images and complex scenes. In recent years, deep learning-based stereo matching methods have emerged to improve matching accuracy. GANet [38] effectively integrates local and global information through the guided aggregation module to enhance the accuracy of disparity estimation. RAFT-Stereo [39] applies a recurrent all-pairs field transform to iteratively optimize cost volumes between all pixel pairs. These deep learning-based methods demonstrate superior adaptability in complex scenes. However, their limitations lie in higher computational complexity, increased hardware requirements, and reduced robustness compared to traditional methods when dealing with noise or low-quality data. Overall, this paper employs the SGBM algorithm for research purposes. To further improve accuracy and real-time performance, this paper proposes an improved multi-threaded optimized SGBM algorithm which introduces multi-threaded parallel computation into the matching cost process for stereo matching.

3. System overview

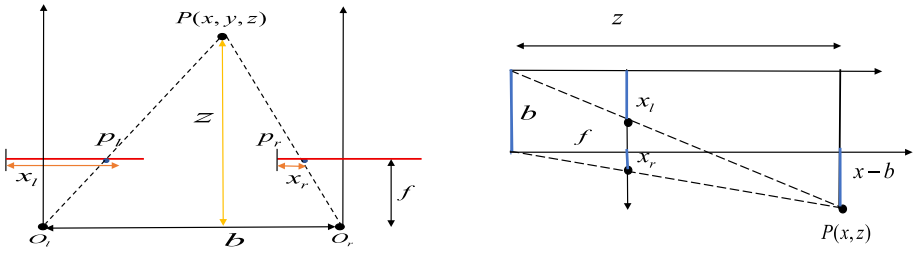
The binocular stereo vision system captures images through two parallel cameras and calculates the disparity between these two images. This process ensures that clear depth information is derived from the correspondence of features in the image. Therefore, an underwater obstacle recognition and distance measurement algorithm based on the binocular stereo vision system is proposed to accurately measure the distance between the camera and obstacles. Ideally, a model of the binocular stereo vision system is illustrated in Fig. 1.

As illustrated in Fig. 1(a), O_l and O_r represent the optical centers of the two cameras, respectively, and the distance between them is considered as the baseline (denoted as b). Point P is a specific location on the object being measured. The corresponding imaging points of point P on the photoreceptors of both cameras are denoted as p_l and p_r, respectively. Here, f represents the focal length of the camera, while z denotes the depth information of the object. The geometric relationship is shown in Fig. 1(b). Based on the principle of triangle similarity, the following relationship is established:

$$\frac{z}{f}=\frac{x}{x_l}=\frac{x-b}{x_r}=\frac{y}{y_l}=\frac{y}{y_r} \tag{1}$$

$$x=\frac{b\times x_l}{x_l-x_r}, z=\frac{b\times f}{x_l-x_r}=\frac{b\times f}{d} \tag{2}$$

where (x_l, y_l) and (x_r, y_r) represent the pixel coordinates corresponding



（a）Binocular camera imaging model (b) Depth computing model.

Fig. 1. Ideal model of the binocular stereo vision system.

to the imaging points p_l and p_r, respectively. f and b can be determined through calibration. Therefore, z can be calculated with the disparity value d.

The underwater binocular vision platform for measuring the distance of dynamic obstacles is depicted in Fig. 2. A binocular camera, an embedded processor (Jason X), and a lighting ring are securely housed within a waterproof housing.

4. The underwater obstacle recognition and distance measurement method

The overall structure of the proposed real-time recognition and distance measurement method for underwater dynamic obstacles is shown in Fig. 3. The camera calibration module provides accurate camera parameters for subsequent measurements. The underwater image enhancement module adopts DCP and ACCLAHE to provide clear images for the obstacle detection module. The YOLOv5s obstacle detection module is used to detect the bounding boxes of obstacles, which serve as input for the obstacle distance measurement module. The obstacle distance measurement module adopts a new stereo matching method based on an improved multi-threaded optimized SGBM algorithm to obtain the obstacle distance information. Ultimately, an image with obstacle labels and distance information is generated in real time.

4.1. Binocular camera calibration module

Before using binocular vision for distance measurement, it is necessary to obtain camera parameters and calibrate the camera. In air, the perspective imaging model is denoted as:

$$z_c\begin{bmatrix}u\\v\\z\end{bmatrix}=\begin{bmatrix}1/dx & 0 & u_0\\0 & 1/dy & v_0\\0 & 0 & 1\end{bmatrix}\begin{bmatrix}f & 0 & 0 & 0\\0 & f & 0 & 0\\0 & 0 & 1 & 0\end{bmatrix}\begin{bmatrix}R & T\\0 & 1\end{bmatrix}\begin{bmatrix}x_w\\y_w\\z_w\\1\end{bmatrix} \tag{3}$$

where z_c is the scale factor, (u, v) is the image coordinate, R and T are the camera rotation matrix and translational vector respectively, and (x_w, y_w, z_w) is the point in the world coordinate. To develop the underwater imaging model, the refraction between water, the waterproof housing and air should be considered.

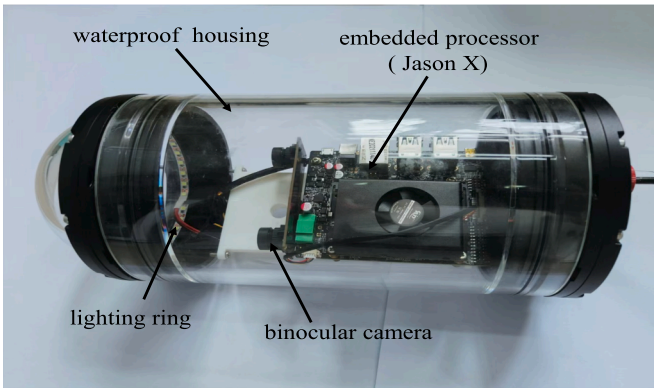


Fig. 2. The underwater binocular vision platform.

According to Snell's law, the following equation can be derived:

$$n_w\sin\theta_w = n_h\sin\theta_h = n_a\sin\theta_a \tag{4}$$

where n_w, n_h and n_a are the refractive indexes of water, the waterproof housing, and air, respectively. θ_w, θ_h, and θ_a are the angles between the incident light and the optical axis in water, the waterproof housing and air, respectively.

Light incident on the waterproof surface undergoes two distinct instances of refraction: one occurs at the interface between water and the waterproof housing, and the other takes place at the boundary between the waterproof housing and air. The underwater imaging model is illustrated in Fig. 4.

As shown in Fig. 4, t1 represents the thickness of the waterproof housing, d_0 indicates the distance from the camera center to the refracting plane, the vectors $(\alpha_w, \beta_w, \gamma_w)^{\mathrm{T}}$ and $(\alpha_a, \beta_a, \gamma_a)^{\mathrm{T}}$ are the direction vectors of the refracted light when it enters water and air, respectively. $P_w(x_w, y_w, z_w)$ is the 3-D spatial point, $M(x_r, y_r, z_r)$ is the refraction point on the waterproof housing, and (u, v) is the point on the imaging plane.

The underwater camera refraction model represents the relationship between the 3-D spatial point and its imaging point. To establish this relationship, the relationship between $(\alpha_w, \beta_w, \gamma_w)^{\mathrm{T}}$ and (x_w, y_x, z_w), as well as the relationship between $(\alpha_w, \beta_w, \gamma_w)^{\mathrm{T}}$ and (u, v), must be determined.

According to the representations of (x_w, y_w, z_w) and $M(x_r, y_r, z_r)$, the relationship between $(\alpha_w, \beta_w, \gamma_w)^{\mathrm{T}}$ and (x_w, y_x, z_w) can be derived as:

$$\begin{bmatrix}x_w\\y_w\\z_w\end{bmatrix}=\begin{bmatrix}x_r\\y_r\\z_r\end{bmatrix}+k\begin{bmatrix}\alpha_w\\\beta_w\\\gamma_w\end{bmatrix} \tag{5}$$

According to the relationship between $M(x_r, y_r, z_r)$ and (u, v), the coordinates of the refraction points can be determined as:

$$\begin{bmatrix}x_r\\y_r\\z_r\end{bmatrix}=\begin{bmatrix}\dfrac{ud}{f}+\dfrac{t_1n_au}{\sqrt{(n_h^2-n_a^2)(u^2+v^2)+n_h^2f^2}}\\[2ex]\dfrac{vd}{f}+\dfrac{t_1n_av}{\sqrt{(n_h^2-n_a^2)(u^2+v^2)+n_h^2f^2}}\\[2ex]t_1+d\end{bmatrix} \tag{6}$$

Finally, introducing (6) into (5), the equation can be derived:

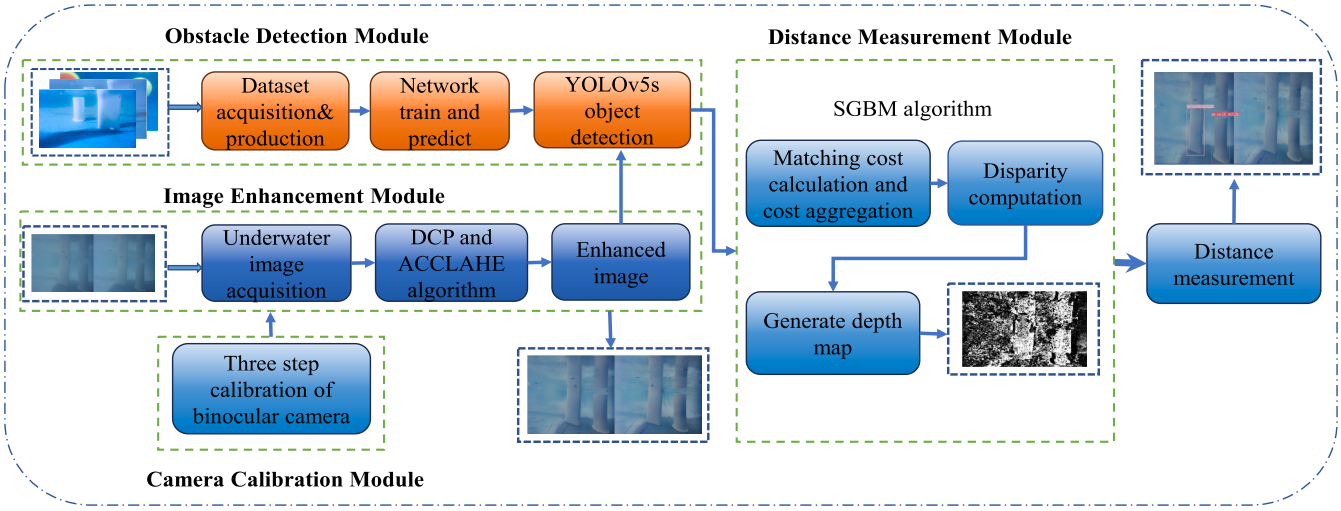


Fig. 3. Underwater obstacle recognition and distance measurement method.

$$k\begin{bmatrix}\alpha_w\\ \beta_w\\ \gamma_w\end{bmatrix}=\begin{bmatrix}x_w\\ y_w\\ z_w\end{bmatrix}-\begin{bmatrix}x_r\\ y_r\\ z_r\end{bmatrix}=\begin{bmatrix}1 & 0 & 0 & -\frac{ud}{f}-\frac{t_1 n_a u}{\sqrt{(n_h^2-n_a^2)(u^2+v^2)+n_h^2 f^2}}\\ 0 & 1 & 0 & -\frac{vd}{f}-\frac{t_1 n_a v}{\sqrt{(n_h^2-n_a^2)(u^2+v^2)+n_h^2 f^2}}\\ 0 & 0 & 1 & -d-t_1\end{bmatrix}\begin{bmatrix}x_w\\ y_w\\ z_w\\ 1\end{bmatrix} \tag{7}$$

The relationship between $(\alpha_a,\beta_a,\gamma_a)^{\mathrm{T}}$ and (u,v) can be obtained:

$$\begin{bmatrix}\alpha_a\\ \beta_a\\ \gamma_a\end{bmatrix}=\frac{1}{\sqrt{u^2+v^2+f^2}}\begin{bmatrix}u\\ v\\ f\end{bmatrix}=\frac{1}{r}\begin{bmatrix}u\\ v\\ f\end{bmatrix} \tag{8}$$

Assuming that the light rays before and after refraction are coplanar, the relationship between $(\alpha_w,\beta_w,\gamma_w)^{\mathrm{T}}$ and $(\alpha_a,\beta_a,\gamma_a)^{\mathrm{T}}$ can be expressed as follows:

$$\begin{bmatrix}\alpha_w\\ \beta_w\\ \gamma_w\end{bmatrix}=k_1\begin{bmatrix}\alpha_a\\ \beta_a\\ \gamma_a\end{bmatrix}+k_2\begin{bmatrix}0\\ 0\\ 1\end{bmatrix},k_1=\frac{n_a}{n_w},k_2=\sqrt{1-\frac{n_a^2}{n_w^2}\sin^2\theta_a}-\frac{n_a}{n_w}\cos\theta_a \tag{9}$$

It follows that:

$$\begin{bmatrix}\alpha_w\\ \beta_w\\ \gamma_w\end{bmatrix}=\frac{n_a}{n_w}\begin{bmatrix}\alpha_a\\ \beta_a\\ \sqrt{n_w^2/n_a^2-1+\gamma_a^2}\end{bmatrix} \tag{10}$$

Based on (8) and (10), the relationship between $(\alpha_w,\beta_w,\gamma_w)^{\mathrm{T}}$ and (u,v) can be derived.

$$\begin{bmatrix}\alpha_w\\ \beta_w\\ \gamma_w\end{bmatrix}=\frac{n_a}{n_w}\begin{bmatrix}\frac{1}{r} & 0 & 0\\ 0 & \frac{1}{r} & 0\\ 0 & 0 & \sqrt{\frac{n_w^2}{n_a^2}-1+\frac{f^2}{r^2}}\end{bmatrix}\begin{bmatrix}u\\ v\\ 1\end{bmatrix} \tag{11}$$

Finally, by utilizing (7) and (11), the physical coordinates of the projection points on the imaging plane can be determined as follows:

$$k\begin{bmatrix}u\\ v\\ 1\end{bmatrix}=\frac{n_w}{n_a}\begin{bmatrix}\frac{1}{r} & 0 & 0\\ 0 & \frac{1}{r} & 0\\ 0 & 0 & \sqrt{\frac{n_a^2}{n_w^2}-1+\frac{f^2}{r^2}}\end{bmatrix}^{-1}\begin{bmatrix}1 & 0 & 0 & -\frac{ud}{f}-\frac{t_1 n_a u}{\sqrt{(n_h^2-n_a^2)(u^2+v^2)+n_h^2 f^2}}\\ 0 & 1 & 0 & -\frac{vd}{f}-\frac{t_1 n_a v}{\sqrt{(n_h^2-n_a^2)(u^2+v^2)+n_h^2 f^2}}\\ 0 & 0 & 1 & -d-t_1\end{bmatrix}\begin{bmatrix}x_w\\ y_w\\ z_w\\ 1\end{bmatrix} \tag{12}$$

Thus, the complete underwater camera imaging model is obtained.

The Zhang calibration method [40] is the most widely employed technique for camera calibration in air. However, due to the light distortion in underwater environments, the Zhang method cannot be used directly for underwater applications. To apply the Zhang calibration method in water, an improved underwater camera calibration model is presented. The proposed method includes three steps:

The first step, calibration in air: Obtaining the intrinsic and extrinsic parameter matrices of the camera.

The second step, calibration after adding a waterproof housing in air: Based on the internal parameters and coordinate transformation matrix calibrated in the first step, the maximum likelihood estimation method is employed to obtain the distortion coefficients of the camera.

$$\begin{bmatrix}(u'-u_0)(x^2+y^2) & (u'-u_0)(x^2+y^2)^2\\ (v'-v_0)(x^2+y^2) & (v'-v_0)(x^2+y^2)^2\end{bmatrix}\begin{bmatrix}k_1\\ k_2\end{bmatrix}=\begin{bmatrix}\widehat{u}-u'\\ \widehat{v}-v'\end{bmatrix} \tag{13}$$

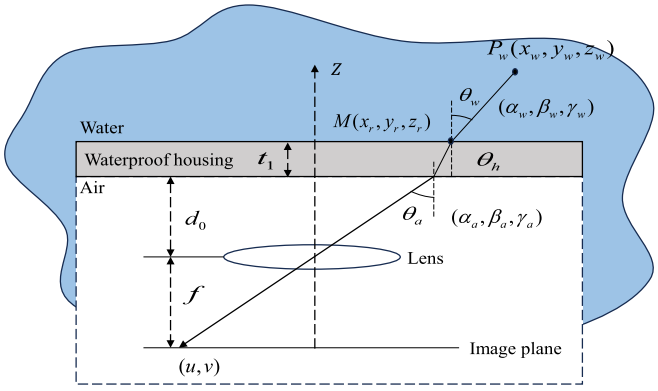


Fig. 4. Underwater imaging model.

where (u_0, v_0) is the pixel translation vector, (u', v') is the coordinate point under ideal conditions without distortion, $(\hat{u}, \hat{v})$ is the coordinate point with real distortion.

The third step, calibration in water: the pixel focal lengths calibrated in air are adjusted to derive the intrinsic parameters and distortion coefficients of the camera in underwater environments.

4.2. Underwater image enhancement module

Due to the absorption and scattering of light, the degradation of underwater images presents a highly complex challenge. To tackle the problem of color distortion in underwater images, the DCP algorithm is employed to estimate atmospheric light and transmission maps for image restoration. However, the restored images still exhibit insufficient local contrast and details in certain areas. To further enhance these aspects, an improved ACCLAHE algorithm, which can automatically determine the number of sub-images based on the content and requirements of the restored image is developed. This algorithm dynamically adjusts clipping limit values according to the local histogram maxima to generate the contrast transformation function for each sub-image. The following sections provide a detailed discussion of both algorithms:

1) DCP: According to Roby's law [41], underwater images I can be expressed as:

$$I(x) = J(x)t(x) + A(1 - t(x)) \tag{14}$$

where x represents the spatial position of the image, $J(x)$ denotes the scene radiance, A represents the atmospheric light, $t(x)$ represents the medium transmission. To enhance underwater images, it is crucial to compute $t(x)$ and A.

The DCP, initially proposed by [9] is commonly used to compute $t(x)$ and A. The dark channel is represented as follows:

$$J^{\text{dark}}(x) = \min_{y \in \Gamma(x)} \left(\min_{\tau \in \{r,g,b\}} \cdot J^{\tau}(y) \right), J^{\text{dark}} \to 0 \tag{15}$$

where y represents the spatial positions of the local image, J^{τ} represents the scene radiance and $\Gamma_{(x)}$ denotes a local patch centered at x.

Taking into consideration the influence of external lighting sources on patch Γ, Ancuti et al. [42] utilize the patch Ψ that is twice the size of Γ. As a result, the local atmospheric light $A_L^{\tau}(x)$ within Ψ is defined as:

$$A_L^{\tau}(x) = \max_{y \in \Psi(x)} \left(\min_{z \in \Gamma(y)} I^{\tau}(z) \right) \tag{16}$$

where $\min_{z \in \Gamma(y)} I^{\tau}(z)$ denotes the lowest intensities of each channel inside Γ. A can be derived by calculating the highest intensities. Once A is obtained, according to (14) and (15), $t(x)$ can be calculated as:

$$t(x) = 1 - \alpha \min_{y \in \Gamma_{(x)}} \left\{ \min_{\tau} \left[\frac{I^{\tau}(x)}{A^{\tau}} \right] \right\} \tag{17}$$

where $\alpha \in (0, 1]$ is the fidelity adjustment factor, which is set to 0.95 in this paper.

Finaly, according to (14). The $J(x)$ is recovered by:

$$J(x) = \frac{I(x) - A}{\max(t(x) \cdot t_0)} + A \tag{18}$$

where t_0 represents the lower limit of the transmission rate.

2) ACCLAHE: The traditional CLAHE method enhances image contrast by dividing the input image into sub-images and applying a user-defined clipping limit to control contrast enhancement. However, traditional CLAHE will suffer from uneven local contrast due to the complex degradation of underwater images. The improved ACCLAHE algorithm presented in this study enables automatic determination of both the number of sub-images ($N \times N$) and clipping limit values according to the image content. The value of clipping limit is defined by the maximum height of bin n in the local histogram associated with each sub-image, and pixels exceeding this limit are then redistributed among all pixels within the sub-image. This approach facilitates precise control over contrast enhancement.

The proposed underwater image enhancement algorithm is articulated as follows:

Algorithm 1: Underwater image enhancement

Input: Image I
Output: Enhanced image
1: Initialization
2: Execute the DCP algorithm
3: Obtain the color restored image
4: Execute the ACCLAHE algorithm
5: Divide the color restored image into $N \times N$ subsections
6: Each subsection follows these steps:
_Compute the histogram for each tile.
_Identify the maximum intensity value within a tile.
_Calculate the clipping limit using the half-interval search method.
_Redistribute the pixels with values exceeding the clipping limit uniformly across all histogram bins in the tile and equalize the histogram.
7: Return the Enhanced image

4.3. YOLOv5s obstacle detection module

To achieve efficient obstacle detection, this paper utilizes the YOLOv5s algorithm, known for its real-time performance and high accuracy in visual recognition tasks. As shown in Fig. 5, YOLOv5s consists of three main components: backbone, neck, and head. The backbone extracts key features, the neck generates feature pyramids to handle object scaling, and the head produces anchor boxes and final output vectors. This design strikes a balance between detection accuracy and speed.

4.4. Distance measurement module

This section presents an enhanced method for estimating the distance of underwater objects using a multi-threaded optimization of the SGBM algorithm. During the matching cost computation of the SGBM algorithm, the MAD-Census [43] transform is employed to replace the traditional SAD cost. Additionally, multi-threaded parallel computation is introduced to enhance real-time performance.

The SGBM algorithm integrates the pixel data along specific directions within a semi-global framework, reducing computational complexity while preserving image details. It involves four key stages: (1) matching cost computation, (2) cost aggregation, (3) disparity computation, (4) disparity refinement. However, the matching cost computation stage is resource-intensive, reducing real-time

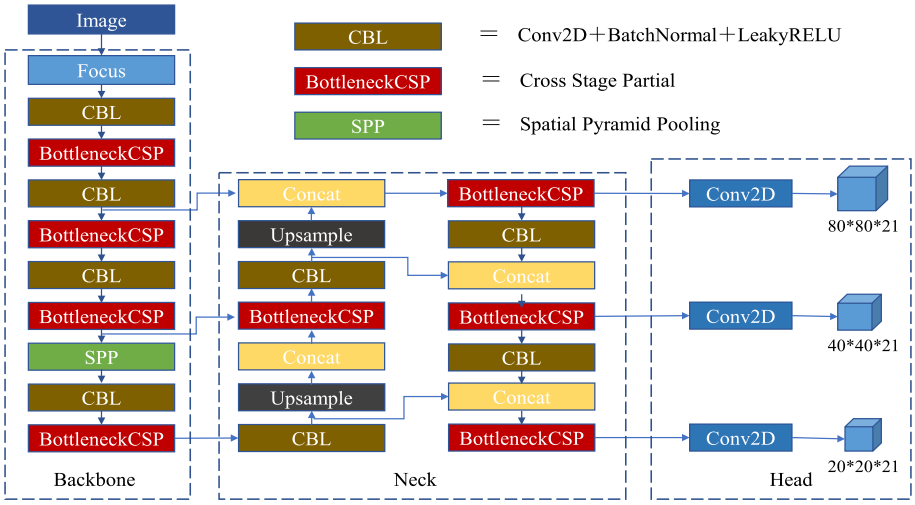


Fig. 5. The architecture of YOLOv5s model.

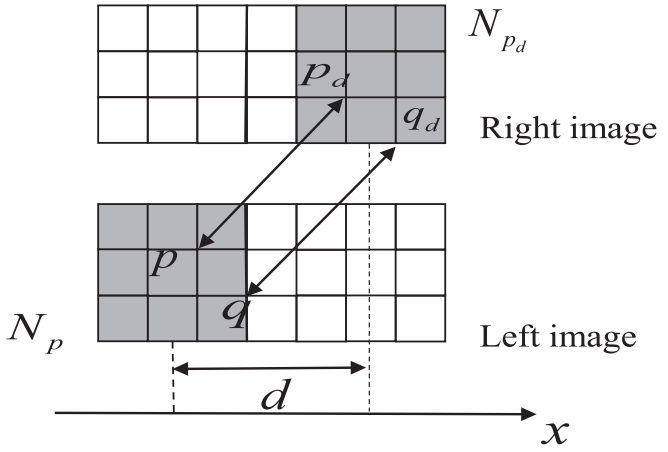


Fig. 6. The detailed process for calculating SAD (N_p is the reference support window and N_{pd} is the target support window when $d_p = d$).

performance. The matching cost comprises two components: the gradient cost, which captures gradient information from the pre-processed image, and the SAD (sum of absolute differences) cost, which is calculated using a sampling-based approach, as shown Fig. 6. The absolute differences between pixel values are computed, and the SAD values are compared within the search range to determine the optimal matching block.

This approach offers fast matching speed in regions with abundant grayscale information and textures. However, it is highly sensitive to image noise and variations in illumination, which can easily lead to mismatches. To tackle this problem, this study introduces an improved SGBM algorithm, which adopts MAD-Census transform for cost computation. The MAD-Census transform, which preserves spatial information of surrounding pixels, demonstrates robustness to changes in illumination. Additionally, the multi-threaded parallel computation is introduced into the matching cost computation to accelerate matching speed.

The Census transform converts the disparity in gray values of adjacent pixels into bit strings and regards the difference between these bit strings as the matching cost of pixel pairs. Fig. 7 shows the two pixels in the left and right graphs within their 3×3 neighborhood window. When a pixel in the surrounding area has a value lower than that of the central pixel, its gray value is assigned to 1. Otherwise, it is assigned to 0. The 8-bit string shown in Fig. 7 can be obtained by connecting the final results in sequence. The value of the bit string is the census transform value $C_{census}(p)$ of this central pixel, which can be calculated as:

$$\xi[I(p), I(q)] = \begin{cases} 0, I(q) \le I(p) \\ 1, I(q) > I(p) \end{cases} \tag{19}$$

$$C_{census}(p) = \underset{q \in N_p}{\oplus} \xi[I(p), I(q)] \tag{20}$$

where $I(p)$ is the gray value of the domain pixel and $I(q)$ is the gray value of the center pixel.

However, the census transform may result in mismatches in repetitive or low-texture areas. Unlike the traditional Census transform, which heavily relies on the stability of the central pixel, the MAD-Census transform computes the average value of the entire region to replace the central pixel during transformation. Furthermore, it uses the absolute difference between region averages as an additional feature to resolve ambiguities in repetitive and low-texture regions, as shown in (21):

$$C_{MAD}(p, d) = \left| \frac{1}{N} \sum_{p \in N_p} - \frac{1}{N} \sum_{q \in N_{p_d}} \right| \tag{21}$$

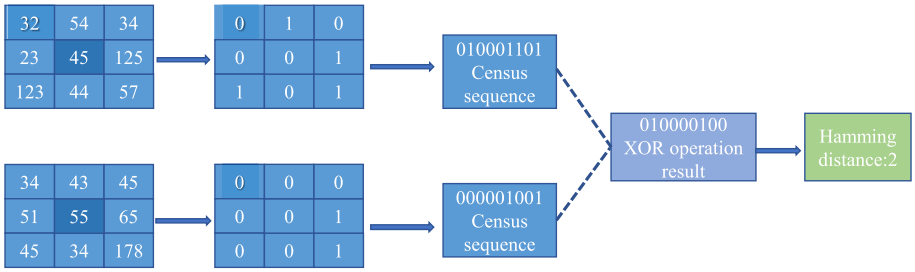


Fig. 7. Census cost calculation process.

The features derived from both transforms are fused to obtain the MAD-Census transform:

$$C_{MAD-Census}(p,d) = \rho(C_{MAD}(p,d),\lambda_{MAD}) + \rho(C_{census}(p,d),\lambda_{Census}) \quad (22)$$

Where c is the cost value, and λ is the fusion coefficient, which indicates the proportional weight of the two features.

From a computational perspective, the Census bit strings are outputted by a 3 × 3 neighborhood window with 8 bits (one byte), which can be conveniently stored in memory and facilitate rapid computation by computer hardware. Additionally, compared to SAD, the MAD-Census transformation is more suitable for integration with the proposed multi-threaded parallel computing approach, since each pixel can be processed independently. Therefore, it significantly enhances computational efficiency.

Finally, based on the disparity value d obtained from the enhanced SGBM algorithm and the associated camera parameters, it is possible to accurately calculate the distance to the obstacle.

5. Experiment

A series of experiments are conducted to evaluate the performance of each module in the proposed recognition and distance measurement method. Detailed information about the experimental platform is presented in Section 5.1. Results for underwater camera calibration, image enhancement, obstacle detection, and stereo matching for distance measurement are discussed in Sections 5.2 to 5.5.

5.1. Experimental platform

The proposed recognition and distance measurement method based on binocular vision is tested on an underwater experimental platform. The dataset used in this experiment is meticulously constructed by the team specifically for this study. To systematically categorize obstacles in underwater images, we classified them into four categories based on their shapes: globosity, cuboid, irregularity, and cylinder. To enhance the model's generalization capabilities, various data augmentation techniques are applied after collection, including rotation, scaling, color adjustment, and noise addition. This process ultimately generates a total of 2600 underwater images for this study. Furthermore, all images are manually annotated by professionals to provide precise bounding boxes and class labels, ensuring high-quality supervisory information during the training process. As shown in Fig. 8, during the experiments, the four types of obstacles trained on the self-built dataset are placed in a water tank measuring 3 m × 2 m × 1 m. A tennis ball is dragged using a tether to simulate the dynamic underwater obstacle. Real-time image acquisition is then conducted using a binocular camera, and the captured underwater images are securely transmitted to an embedded computer housed within the waterproof enclosure. Throughout the experiment, when the computer receives images from the camera, it executes the recognition and distance measurement method to determine the distance from the obstacles to the camera. The measurement results are subsequently transferred to the monitor through a cable that connects the waterproof housing and the monitor.

5.2. Camera calibration results

To validate the proposed three-step calibration method for the binocular camera, three separate calibration experiments are conducted under distinct conditions: in air, in air with the waterproof housing, and in water. These experiments yield the internal parameter matrices, distortion coefficients, and external parameter matrices of the camera. The accuracy of the calibration process is evaluated by analyzing the reprojection error, which quantifies the difference between the observed image points and the corresponding points projected from the calibration model. A lower reprojection error indicates higher calibration precision, demonstrating the reliability of the estimated camera parameters.

First step: The calibration experiment in air involves moving a chessboard in front of the camera at distances between 30 cm and 60 cm.

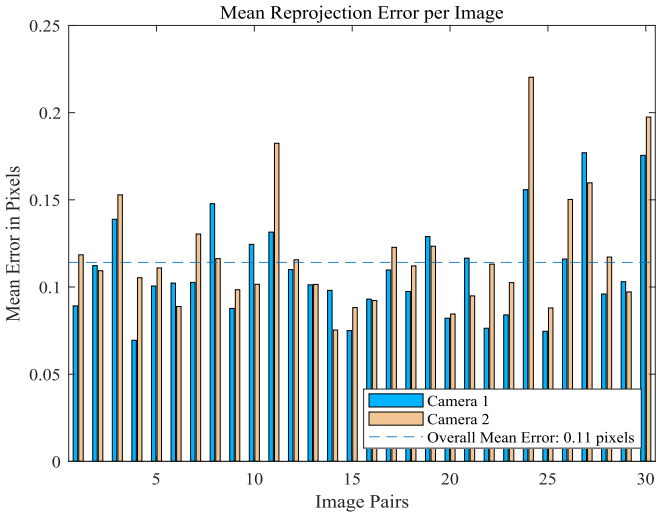


Fig. 9. The distribution of reprojection errors in air.

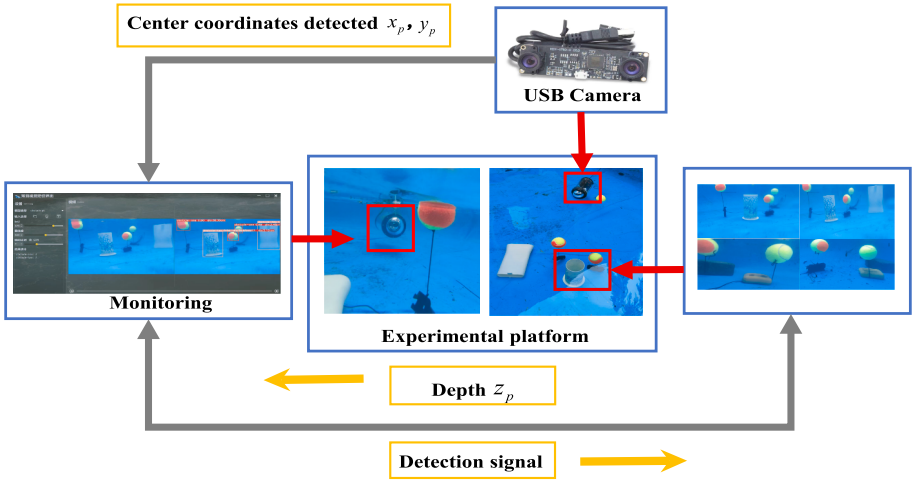


Fig. 8. Experimental platform.

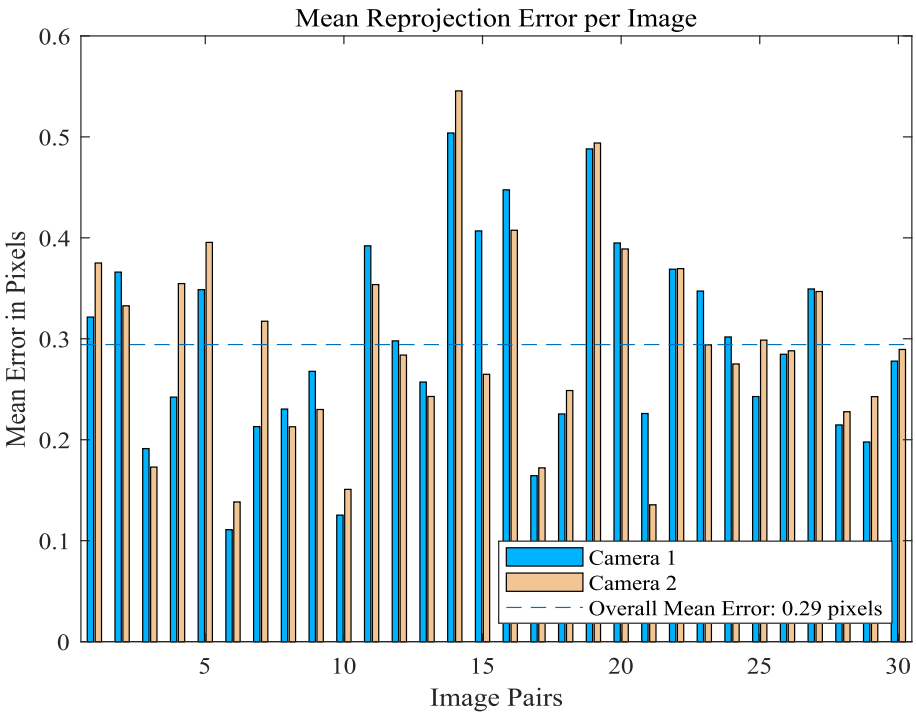


Fig. 10. The distribution of reprojection errors in air with the housing.

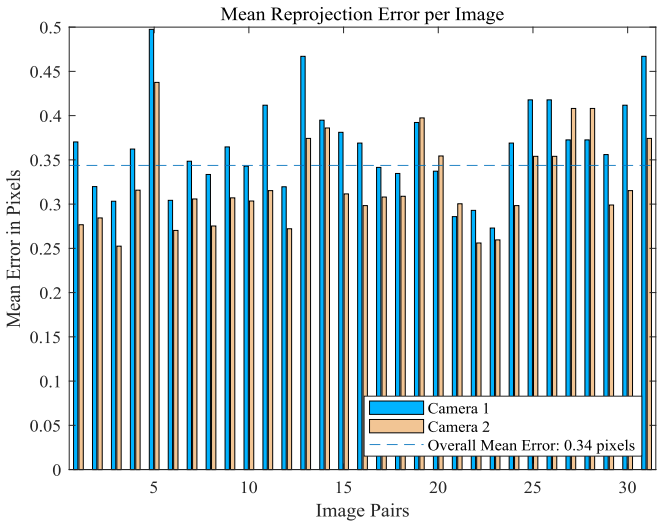


Fig. 11. The distribution of reprojection errors in water.

A total of 30 image sets of the calibration board in various poses are processed using the MATLAB calibration program to detect all the corners. The distribution of reprojection errors for each image is illustrated in Fig. 9.

Second step: To derive the influence of the waterproof housing on the calibration result, the camera is fixed inside the housing to capture the chessboard. The distribution of reprojection errors for each image in air with the housing is illustrated in Fig. 10.

Third step: The third experiment is carried out in water. The distribution of reprojection errors for each image in water with the housing is depicted in Fig. 11.

As shown in Fig. 9, during calibration in air, ambient light conditions remain relatively stable, and light is not refracted by additional media. This scenario represents the ideal calibration environment, resulting in minimal reprojection error (typically less than 0.3 pixels). The calibration produces a reprojection error of 0.11 pixels, which is considered indicative of high calibration accuracy. In contrast, as shown in Fig. 10, when a waterproof housing is used in air, light has to pass through the housing window, introducing refraction effects that slightly alter the light path. This leads to some deviation from the optimal calibration results, with the measurement recording a reprojection error of 0.29 pixels. During underwater calibration, light has to traverse multiple layers of refraction: first through air, then through the housing shell, and finally through water. This results in larger errors compared to the previous scenarios. As shown in Fig. 11, a reprojection error of 0.34

Table 1
The intrinsic parameters of the left and right camera.

Parameter	Air-L	Air-R	Plastic-L	Plastic-R	Water-L	Water-R
f_x	725.7247	728.4319	737.1322	739.2774	962.4309	969.7221
f_y	725.4356	728.5826	736.9749	739.1462	963.0461	970.3343
c_x	604.9829	632.4986	605.5034	634.4265	609.0834	641.0056
c_y	325.1856	320.0706	324.5903	313.6753	16.5391	318.4419
p_1	0.0005	0.0008271	0.0031	0.0013	−0.0042	0.0018
p_2	−0.0020	0.0005514	0.0003	0.0021	−0.0026	0.0082
k_1	0.0694	0.0687	0.0700	0.0965	0.5077	0.4727
k_2	0.0496	0.0754	0.0553	−0.0777	0.5351	0.9300
k_3	−0.2800	−0.3323	−0.2775	−0.1108	−2.0719	−3.1883

pixels is measured in this study. It is widely accepted that a reprojection error below 0.5 pixels indicates high accuracy in this context. Therefore, the calibration results obtained in this study can be considered highly accurate within this range of precision.

The intrinsic parameters of the left and right camera generated from the calibration experiments are recorded in Table 1, where f_x and f_y are the focal lengths of the left and right camera respectively, c_x and c_y are the location of the principal point (center of light) within the pixel coordinate system, p_1 and p_2 are the tangential distortion coefficients, k_1, k_2 and k_3 are the radial distortion coefficients.

The external parameters of the underwater binocular camera can be expressed by the rotation matrix R and translational vector T:

$$R = \begin{bmatrix} 0.9999 & -0.0009 & -0.0119 \\ 0.0008 & 1.0000 & -0.0068 \\ 0.0119 & 0.0068 & 0.9999 \end{bmatrix} \tag{23}$$

$$T = [\,-249.1625 \quad -0.1728 \quad 0.0028\,] \tag{24}$$

It can be observed from Table 1 that the waterproof housing (3 mm plastic) has minimal impact on the focal length and translation parameters of the camera, while it primarily affects the distortion coefficients. Furthermore, the underwater environment also influences the distortion coefficients and focal length of the camera. The focal lengths of the air model and underwater model are listed in Table 2. It can be calculated that the focal length of the underwater model shows a ratio of 1.33 compared to that of the air model, which is consistent with the conclusion presented in [44]. The experimental results indicate that an accurate calibration model of the underwater binocular camera can be derived using the proposed three-step calibration method.

5.3. Underwater image enhancement results

In order to illustrate the effectiveness of the proposed image enhancement method, a comparison is made between the proposed algorithm and six advanced image processing algorithms: ACDC[45], Retinex [13], CLAHE [12], DCP [9], MSPE [46] and La-Net [47].

A suite of non-referenced evaluation metrics is adopted for an objective assessment of enhanced image quality without the need for reference images. The non-referenced evaluation metrics employs in this study are listed below.

- PSNR (Peak Signal-to-Noise Ratio) [48]: PSNR is the ratio between the maximum signal power and the noise power impacting the signal. Higher PSNR values indicate better image quality. Generally, a PSNR value above 40 dB is considered excellent. Given a clean image G and a noisy image K of size $m \times n$, the MSE (Mean Square Error) is defined as:

$$MSE = \frac{1}{mn}\sum_{i=0}^{m-1}\sum_{j=0}^{n-1}[G(i,j) - K(i,j)]^2 \tag{24}$$

The PSNR is then computed as:

$$PSNR = 10 * \log_{10}(\frac{MAX_I^2}{MSE}) \tag{25}$$

Where MAX_I^2 represents the maximum achievable pixel value of the image.

Table 2
Comparison of focal lengths between the air model and underwater model.

	Focal length	Air model	Underwater model	ratio
Left camera	f_x	725.7247	962.4309	1.3261
	f_y	725.4356	963.0461	1.3275
Right camera	f_x	728.4319	969.7221	1.3312
	f_y	728.5826	970.3343	1.3318

- SSIM (Structural Similarity) [49]: SSIM is a full-reference metric for image quality assessment that measures the similarity between two images based on brightness, contrast, and structure. A higher SSIM value signifies better image quality, with a maximum value of 1. The SSIM is defined as:

$$SSIM = \frac{(2\mu_x\mu_y + C1)(2\sigma_{xy} + C2)}{(\mu_x^2 + \mu_y^2 + C1)(\sigma_x^2 + \sigma_y^2 + C2)} \tag{26}$$

Where μ_x and μ_y denote the average values of two images, σ_x and σ_y signifies the total number of pixel values, σ_{xy} refers to the covariance matrix of the two images, $C1$ and $C2$ are constants.

- UIQM (Underwater Image Quality Measurement) [50]: UIQM is a no-reference quality index for underwater images. It is computed as a linear combination of three key metrics: Underwater Image Colorfulness Measure (UICM), Underwater Image Sharpness Measure (UISM), and Underwater Image Contrast Measure (UIConM). The UIQM is defined as follows:

$$UISM = \sum_{c=1}^{3}\lambda_c EME(grayscaleedge), EME = \frac{2}{h_1 h_2}\sum_{l=1}^{h_1}\sum_{h=1}^{h_2}\log(\frac{I_{\max,h,l}}{I_{\min,h,l}}) \tag{27}$$

$$UIConM = \frac{1}{h_1 h_2}\sum_{l=1}^{h_1}\sum_{h=1}^{h_2}\frac{I_{\max,h,l} - I_{\min,h,l}}{I_{\max,h,l} + I_{\min,h,l}}\log(\frac{I_{\max,h,l} - I_{\min,h,l}}{I_{\max,h,l} + I_{\min,h,l}}) \tag{28}$$

$$UICM = -0.0268\sqrt{\mu_{a,RG}^2 + \mu_{a,YB}^2} + 0.1586\sqrt{\sigma_{a,RG}^2 + \sigma_{a,YB}^2} \tag{29}$$

$$UIQM = c_1 \times UICM + c_2 \times UISM + c_3 UIConM \tag{30}$$

Where λ_c represents the three channels, EME serves as an indicator for evaluating image distortion. The image is partitioned into $h_1 \times h_2$ blocks, where $(I_{\max,h,l})$ and $(I_{\min,h,l})$ denote the maximum and minimum gray values within each block area, respectively. RG and YB represent two contrasting color components, μ_a signifies the mean under truncated statistics, while σ_a indicates the variance of these statistics, c_1, c_2, and c_3 are the corresponding weighting coefficients.

- UCIQE (Underwater Color Image Quality Evaluation) [51]: UCIQE is a metric for evaluating underwater image quality, focusing on three main attributes: chroma, saturation, and contrast. It is computed as a linear combination of these measures. A higher UCIQE value indicates better balance of these attributes and overall higher image quality. The UCIQE is defined as follows:

$$UCIQE = c_1 \times \sigma_c + c_2 \times con_l + c_3 \times \mu_s \tag{31}$$

where σ_c represents the standard deviations of chrominance, con_l denotes brightness contrast, μ_s is average saturation.

The experimental results are shown in Fig. 12. As illustrated in Fig. 12(b), ACDC effectively mitigates color deviation. However, it results in low saturation and a lack of vibrancy. As shown in Fig. 12(c), the Retinex method improves brightness and reduces noise but causes a washed-out appearance due to fog effects. Fig. 12(d) demonstrates that the CLAHE method is less effective under low-light and backscattering conditions, as it primarily targets local contrast. The DCP method, as shown in Fig. 12(e), fails to enhance luminosity or contrast and may exacerbate color biases due to the degradation caused by underwater conditions. As depicted in Fig. 12(f), MSPE introduces an additional red bias. Furthermore, as shown in Fig. 12(g), La-Net enhances the overall image quality but compromises certain details. In contrast, as shown in Fig. 12(h), the proposed approach significantly enhances image details and brightness while reducing noise and outperforming other methods. To further demonstrate the effectiveness of the proposed image enhancement algorithm, the images presented in Fig. 12 also show results from obstacle detection. As shown in Fig. 12(a)–(h), the average accuracies for all detected obstacles are recorded as follows: 0.89, 0.29,

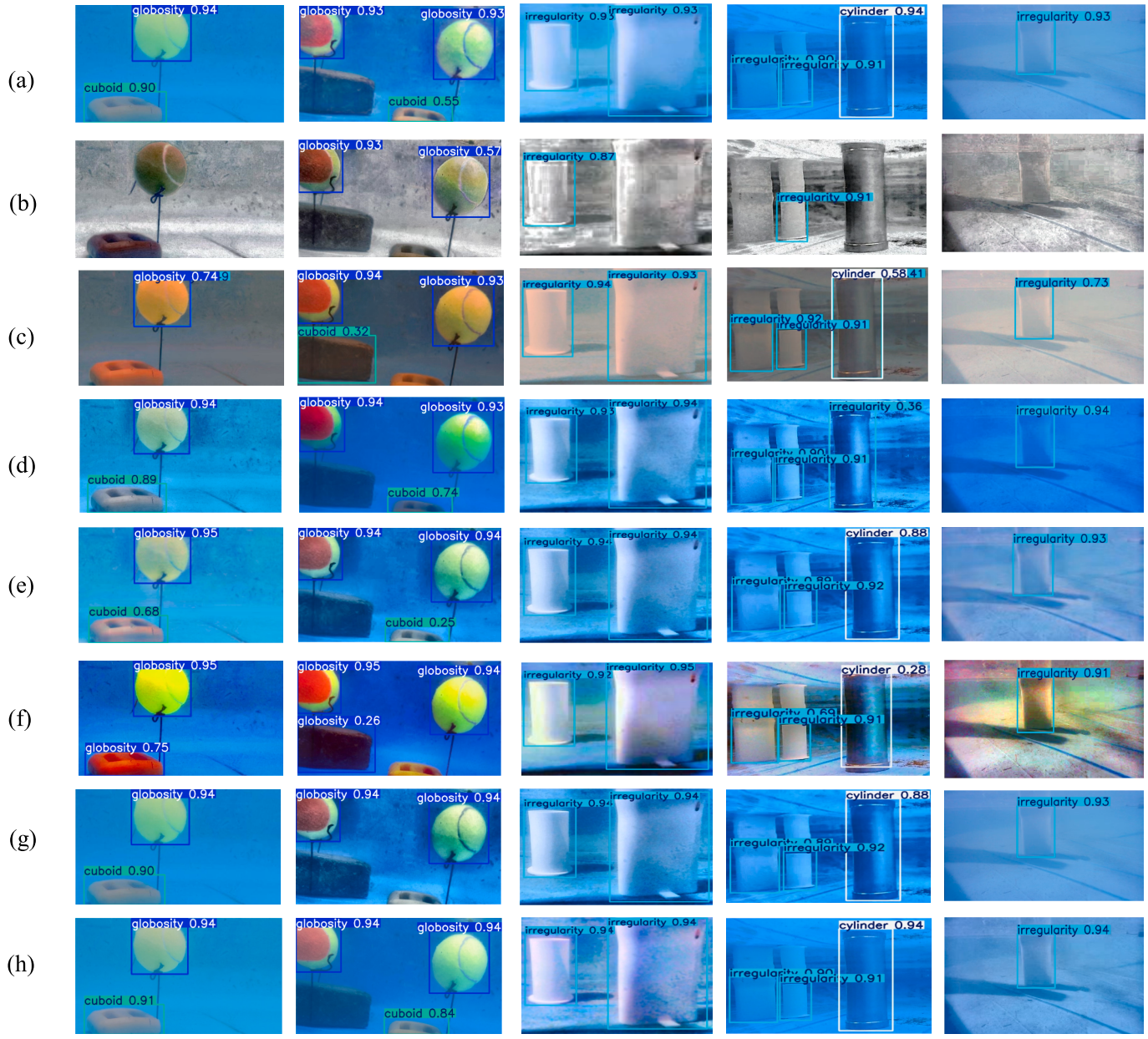


Fig. 12. Comparison of image enhancement algorithms. (a) Raw image, (b) ACDC [45], (c) Retinex [13], (d) CLAHE [12], (e) DCP [9], (f) MSPE [46], (g) La-Net [47], (h) Proposed.

Table 3
The results of seven evaluation metrics.

Methods	PSNR	SSIM	UCIQE	UICM	UIConM	UISM	UIQM
Raw image	24.1207	0.0753	0.1643	2.0715	0.0789	3.4143	1.1143
ACDC	37.4201	0.5817	0.0279	−5.7843	0.3423	1.2148	−1.8917
Retinex	25.7314	0.5652	0.0143	−3.1415	0.7896	2.1423	−1.4216
CLAHE	33.2338	0.6318	0.2143	4.1328	0.3415	7.1423	4.1419
DCP	22.7815	0.5513	0.6513	7.2739	0.1327	7.2913	5.7125
MSPE	24.3705	0.7426	0.6203	5.2279	0.6974	3.5423	3.5937
La-Net	50.3217	0.6872	0.3445	6.1427	0.6892	8.1428	5.7718
proposed	48.6709	0.6634	0.6709	6.1109	0.7314	15.1127	7.6313

0.72, 0.85, 0.83, 0.66, 0.83, and 0.92, respectively. Notably, the accuracy achieved by the proposed algorithm is 0.92, and all targets can be accurately identified. In contrast, other algorithms exhibit lower detection accuracies. Moreover, their image enhancement effects are insufficient for effective object detection. In summary, the proposed image enhancement algorithm demonstrates the most significant improvement in detection accuracy among all evaluated algorithms.

Table 3 shows the evaluation results for the PSNR, SSIM, UCIQE, UICM, UIConM, UISM, and UIQM metrics across 50 images that feature the four types of target objects. The experimental results demonstrate

that the proposed algorithm outperformed others in both the UIConM and UISM metrics, indicating a significant advantage in preserving image details and enhancing contrast. In terms of color performance, the DCP algorithm achieves the highest UICM value, highlighting its proficiency in color reproduction. However, due to noise interference, this improvement in color restoration does not substantially enhance overall image quality. In conclusion, the proposed algorithm offers notable improvements in overall image quality, particularly in maintaining image details.

5.4. Obstacle detection results

The self-built dataset comprises 2600 images, encompassing both single-object and multi-object instances to ensure data diversity and comprehensiveness. To systematically process the obstacles in the underwater images, we classified them into four categories based on their shapes: globosity, cuboid, irregularity, and cylinder. Specifically, globosity refers to obstacles that are spherical or nearly spherical in shape, cuboid refers to those resembling a cuboidal or rectangular shape, irregularity encompasses obstacles with non-standard shapes that defy easy categorization into conventional geometric forms, and cylinder refers to obstacles with a cylindrical shape. During the data collection process, four types of obstacles are placed in the experimental pool to simulate the complex conditions of a real underwater environment. Additionally, certain obstacles are arranged to occlude one another to increase the challenges associated with the detection task. During image acquisition, a binocular camera is employed to capture images under varying water quality conditions, specifically clear and turbid environments. To obtain diverse image data, the camera is moved through multiple angles, capturing images from various perspectives and distances. Finally, the self −built dataset of 2600 images is divided into a training set, validation set and a testing set in a ratio of 6: 2: 2. The dataset has been open-sourced on GitHub. The open source link:https://github.com/penghai0222/Underwater-obstacle-data-set.git.
Table 4 shows the number of instances of each class in the training set, validation set and test set. The YOLOv5s model is trained on a laptop computer. The experimental environment on the server includes a Windows 10 operating system with an i5-8265U CPU, an NVIDIA GeForce 1650 Ti GPU, PyTorch version 1.12 with CUDA 11.6 support, as well as Torchvision version 0.9 and Python version 3.8.

Parameter sizes and FLOPs are selected as evaluation metrics to assess computational requirements and complexity. Additionally, FPS is chosen as an index for assessing model speed. For accuracy evaluation, *mAP* and *F*1-score (*F*1) are used as metrics. The formulas are presented as follows:

$$Precision = \frac{TP}{TP + FP} \tag{32}$$

$$Recall = \frac{TP}{TP + FN} \tag{33}$$

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{34}$$

$$mAP = \frac{\sum_{q_0=1}^{Q_0} \sum_{j_0=1}^{J_0} P(\varepsilon) \Delta R(\varepsilon)}{Q_0} \tag{35}$$

Table 4
Number of instances of each class on the training set, validation set and test set.

Dataset	Training set	Validation set	Testing set
globosity	1570	459	379
cuboid	540	243	210
irregularity	1279	189	143
cylinder	485	223	206

where Q_0 represents the number of categories, J_0 represents the number of samples, ε represents a certain sample, *TP* represents a correct detection box, *FP* represents a false check box, and *FN* represents a missed check box.

To effectively demonstrate the real-time performance and accuracy of YOLOv5s in obstacle detection, comparative experiments are conducted using typical real-time models: YOLOv3-tiny [22], YOLOv8s [26], YOLOv9s [27], YOLOv10s [28], and YOLOv11s [29] on the same dataset. All the images are resized to 640 × 640 pixels. The parameters for each model are documented in Table 5, while the experimental results are presented in Table 6. Although the YOLOv9s model exhibits superior accuracy, with *F*1 and mAP50 scores of 99.57 % and 99.62 %, respectively, it is unsuitable for real-time detection due to its high computational complexity and slow inference speed. In contrast, the YOLOv5s model used in this study shows slightly lower accuracy than YOLOv9s, but it achieves the fastest inference speed at 162 FPS, which is much higher than that of YOLOv9s. Compared to YOLOv3-tiny, YOLOv8s, YOLOv10s, and YOLOv11s, YOLOv5s exhibits significant advantages in terms of detection accuracy, computational complexity, and detection speed.

To more effectively demonstrate the superiority of the YOLOv5s model, inference tests were conducted on the test set using the trained model. The results of these tests are shown in Fig. 13. In scenarios where objects were obscured and blurred, YOLOv5s achieved an identification accuracy of 87 %, whereas other algorithms frequently encountered category recognition errors and failed to accurately identify target objects in obstacle detection tasks. In summary, the YOLOv5s model utilized in this study exhibits superior performance in terms of detection accuracy and speed, making it highly suitable for real-time obstacle detection applications.

5.5. Obstacle distance measurement results

The accuracy of stereo matching is reflected in the disparity map generated during the process, which comprises disparity values from all pixel points. The real-time performance can be assessed by measuring the duration required for the entire matching procedure. To better demonstrate the superiority of the proposed algorithm, BM [35], SGBM [36], RAFT-Stereo [39], IGEV-Stereo [52], and ASTR [33] are applied to the same set of images processed by image enhancement algorithms. Fig. 14 shows the disparity maps generated by these methods, and Table 7 lists their stereo matching times. Fig. 14(b) reveals that the depth map generated by the BM algorithm may present mismatches in regions with sparse textures or low contrast. This is because BM focuses on local matching, leading to inferior performance. Fig. 14(c) presents the disparity map generated by the original SGBM algorithm. It is clear that effective contour matching occurs only in high-contrast regions, while matching details are still lacking. Fig. 14(d) demonstrates the disparity map generated by SGBM on the enhanced image. Compared to Fig. 14(c), the enhanced image enables more precise matching, as it can distinctly differentiate the target object. Nevertheless, due to image degradation, numerous mismatches still occur. Moreover, as illustrated in Table 7, the matching speed of SGBM, both before and after image enhancement, is slower than that of BM. In Fig. 14(e)−(g), it can be observed that the disparity maps generated by RAFT-Stereo, IGEV-Stereo, and ASTR effectively separate the target object from the background. However, when confronted with small target objects with sparse textures (such as small balls), these methods result in mismatches. Fig. 14(h) presents the results of the proposed algorithm. In contrast to Fig. 14(b)-(f), the proposed algorithm offers improved details when targeting areas are characterized by low texture or irregular shapes. Therefore, the proposed algorithm offers accurate matching capabilities for small objects with sparse textures. Moreover, as shown in Table 7, the introduction of multi-threaded parallel computing significantly enhances the matching speed of the proposed algorithm. The matching process can be completed in only 0.3534 s, making it the fastest among

Table 5
The parameters for each model.

Methods	Learning rate	Optimizer	Momentum	Weight decay	Epochs	Batch size
YOLOv3-tiny	1e-2	SDG	0.937	5e-4	100	16
YOLOv8	1e-2	SDG	0.937	5e-4	100	16
YOLOv9	1e-2	SDG	0.937	5e-4	100	16
YOLOv10	1e-2	SDG	0.937	5e-4	100	16
YOLOv11	1e-2	SDG	0.937	5e-4	100	16
YOLOv5s	1e-2	SDG	0.937	5e-4	100	16

Table 6
The detection results of different models.

Methods	P(%)	R(%)	F1	mAP50 (%)	Params (M)	FLOPs (G)	FPS
YOLOv3-tiny	99.45	99.33	99.19	99.57	9.52	17.3	65
YOLOv8s	99.36	99.57	99.55	99.45	9.83	27.5	153
YOLOv9s	99.42	99.42	99.57	99.62	6.19	25.6	97
YOLOv10s	99.15	98.79	98.67	99.47	8.04	20.6	115
YOLOv11s	99.50	99.12	99.38	99.45	9.41	21.3	146
YOLOv5s	99.09	99.45	99.23	99.57	7.03	16.0	162

all compared methods. Consequently, the proposed algorithm strikes an optimal balance between matching accuracy and speed.

By utilizing the generated disparity maps, the relative distances between the camera and obstacles are calculated. For each obstacle, 30 measurements are taken to obtain the average distance for each algorithm. The average error is chosen as the evaluation metric for assessing the accuracy of distance measurements [53,54]. The distances between the camera and obstacles for different algorithms are shown in Table 8. The actual distance is measured manually before each test. As shown in Table 8, the average error of the original SGBM algorithm is 29.6 %. After image enhancement, the average error decreases to 19.4 %. When employing the proposed algorithm, the error is further reduced to 11.7 %. Notably, the average error of the proposed algorithm is much lower when compared to other algorithms. The experimental results suggest that the proposed method holds significant potential for improving the accuracy of depth estimation in underwater environments and highlights the effectiveness of image enhancement techniques in refining distance measurement precision.

6. Conclusion

This paper proposes a rapid and accurate method for recognizing and measuring the distances to dynamic underwater obstacles using binocular vision. The Zhang calibration method is improved by incorporating a new underwater camera refraction model. Additionally, an image enhancement technique combining DCP and ACCLAHE is employed to enhance contrast and details while mitigating color distortion. An enhanced multi-threaded SGBM method is developed for measuring distances to underwater obstacles. Comparative experiments demonstrate that the proposed method achieves the highest precision in measuring underwater obstacle distances. Key advantages include significant improvements in contrast and color balance, a substantial reduction in measurement errors in underwater environments, and notable real-time performance. However, the proposed method has certain limitations. First, the measurement range is constrained by lighting conditions and light attenuation in underwater environments. Second, the effectiveness of the binocular vision system is influenced by water clarity. In highly turbid or murky water, the accuracy of distance measurement and obstacle recognition may deteriorate. While image enhancement techniques improve contrast, they cannot fully

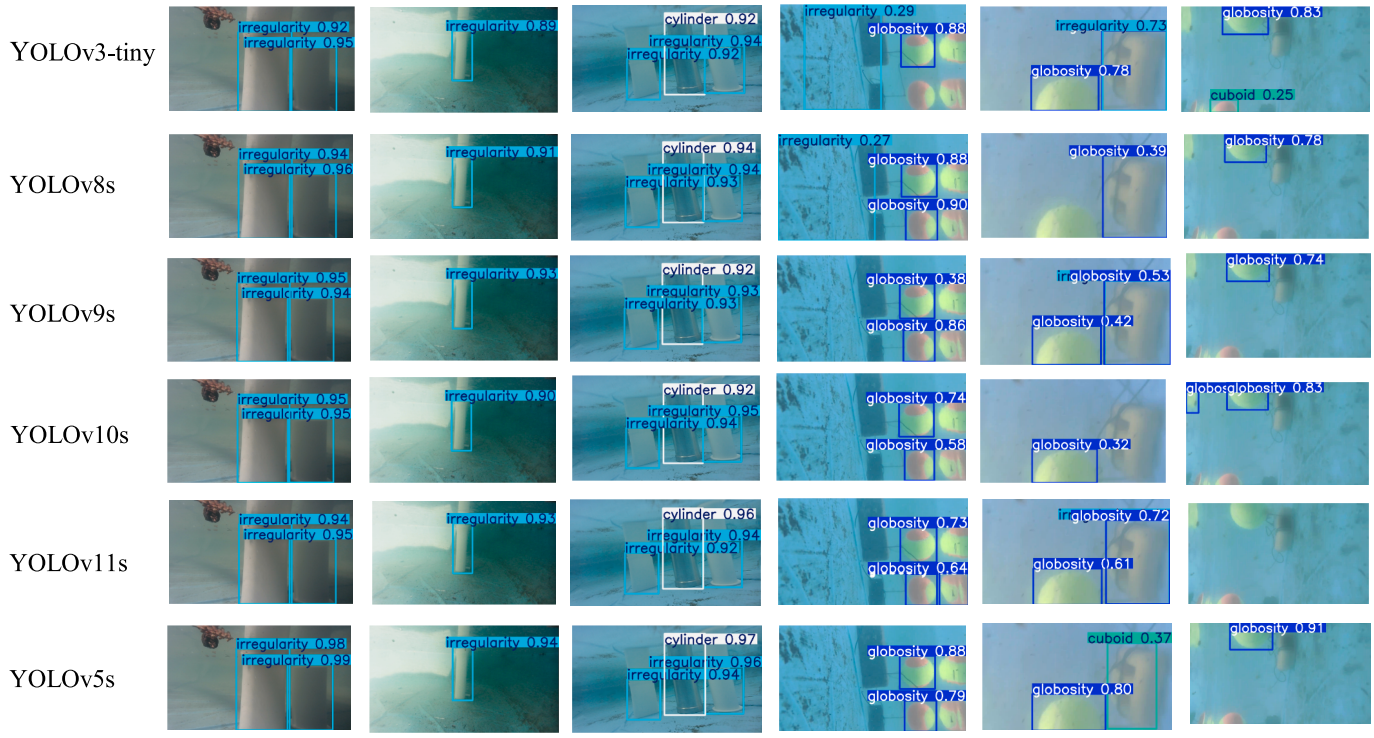


Fig. 13. Detection results for different models.

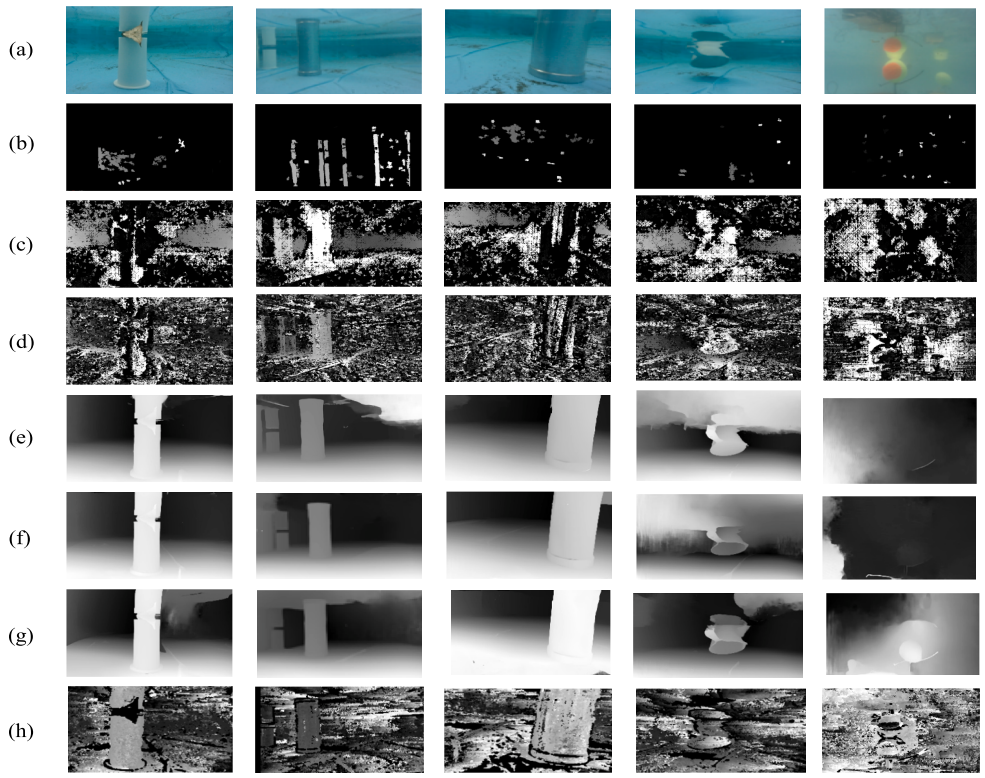


Fig. 14. Evaluations of various stereo matching methods (a) Original image, (b) BM [35],(c) SGBM [36] with original images, (d) SGBM with enhanced images, (e) RAFT-Stereo [39], (f) IGEV-Stereo [52], (g) ASTR [33],(h) proposed.

Table 7
Comparison of time consuming for different stereo matching methods.

Image group	Globosity (s)	Cuboid (s)	Irregularity (s)	Ball (s)	Cylinder (s)	Average time (s)
BM	0.273	0.528	0.431	0.451	0.151	0.366
SGBM(original image)	0.978	1.230	0.948	0.754	0.447	0.871
SGBM	0.832	1.144	0.887	0.612	0.354	0.765
RAFT-Stereo	1.723	1.324	1.791	1.675	1.322	1.571
IGEV-Stereo	0.892	1.231	1.579	1.647	1.792	1.228
ASTR	1.238	1.297	0.747	1.489	0.391	1.032
Proposed	0.248	0.497	0.357	0.379	0.286	0.353

Table 8
Average measurements of obstacle distances by different methods.

Methods	Globosity (mm)	Cuboid (mm)	Irregularity (mm)	Ball (mm)	Cylinder (mm)	Error (%)
Actual Distance	500	800	1000	1200	1500	
BM	900	1150	1320	1450	1800	39.3
RAFT-Stereo	600	1000	900	1100	1600	13.3
IGEV-Stereo	650	560	920	1300	1300	16.4
ASTR	620	720	1100	1400	1615	13.8
SGBM(original image)	920	1000	1170	1120	1750	29.6
SGBM(enhance image)	770	950	1080	1250	1660	19.4
Proposed	370	640	950	1216	1600	11.7

compensate for the degradation caused by poor water quality. Overall, the proposed algorithm effectively measures the distances to dynamic underwater objects. Future research will focus on introducing multi-sensor fusion methods that combine data from binocular vision, sonar, and other sensors. Sonar provides supplementary information in low-visibility and turbid waters. Therefore, multi-sensor fusion based method can improve the robustness and accuracy of underwater obstacle detection and distance measurement in highly turbid or murky waters.

CRediT authorship contribution statement

Qi Chen: Writing – review & editing, Funding acquisition, Conceptualization. **Hui Liu:** Writing – original draft, Software, Methodology. **Wenyang Gan:** Visualization, Validation, Resources, Investigation, Formal analysis, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported by the National Nature Science Foundation of China under Grants 52471336, 52127813 and 52101362

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.measurement.2025.117329.

Data availability

Data will be made available on request.

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